Exploratory Analysis of Student Classroom Behaviour, Lifestyle Habits, and Their Impact on Academic Performance and Stress

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Abstract—
This study explores the relationship between student classroom behavior, lifestyle habits, and academic outcomes. A dataset consisting of 66 student responses was collected through a structured questionnaire capturing demographic information, study habits, sleep patterns, screen usage, and classroom engagement. The analysis includes data preprocessing, descriptive statistics, exploratory data analysis, and hypothesis testing using Chi-square tests. The results indicate that classroom participation and seating preferences influence attention levels, while lifestyle factors such as screen time and sleep duration significantly impact academic stress. The findings highlight the importance of healthy study habits and classroom engagement in improving academic performance.

Keywords—*Student Behaviour, Academic Performance, Exploratory Data Analysis, Chi-Square Test, Educational Data Mining*

1. INTRODUCTION

There have been many discussions and differing opinions among students from various colleges regarding the factors that influence academic performance and student stress. Some believe that classroom behaviors such as active participation, seating preference, phone usage during lectures, and attention levels can significantly affect academic performance and overall CGPA. Similarly, lifestyle habits such as assignment management, study routines, screen time, and sleep patterns are often considered important factors influencing the academic stress experienced by students.

In this study, we aim to explore whether these commonly held assumptions are supported by data through Exploratory Data Analysis (EDA). Data for this analysis was collected using an online survey designed to capture information related to students' classroom behavior, study habits, digital device usage, sleep patterns, and perceived academic stress levels. The goal of this study is to gain insights into how these factors may influence students' academic experiences and overall well-being.

2. RESEARCH QUESTIONS

This study is guided by the following research questions:

1. **Does seating preference and classroom participation influence academic performance and engagement?** In this analysis, we examine whether classroom seating location, participation in discussions, and study habits are associated with variations in students' academic performance, particularly their CGPA and attention levels during lectures.
2. **How do sleep patterns, screen usage, and assignment management habits influence academic stress among students?** This question explores whether lifestyle-related factors such as sleep duration, digital device usage, and assignment preparation habits are associated with differences in the academic stress levels reported by students.

By analyzing EDA and descriptive statistics results, the study aims to identify patterns that may help better understand factors influencing students' academic experiences.

This report presents a study on how classroom behaviour, Lifestyle Habits will Impact there Academic Performance and Stress Levels.

3. METHODOLOGY (DATA COLLECTION METHODS)

To conduct this study, a primary dataset was collected through an online survey. **Data collection plays an important role** in experiments involving exploratory data analysis, as the quality and relevance of the dataset directly affect the insights obtained from the analysis.

A structured questionnaire was designed using Google Forms to collect responses related to students' classroom behavior, study habits, sleep patterns, screen usage, and academic stress levels. The survey consisted of multiple questions that included both **quantitative variables** (such as study hours, phone usage during lectures, and CGPA) and **qualitative variables** (such as seating preference, participation frequency, and assignment preparation habits).

For distributing the survey, the **snowball sampling technique** was used. In this method, the survey link was initially shared with a small group of students through platforms such as WhatsApp. These participants were requested to fill out the survey and further share it with their peers. This approach helped in reaching a wider group of respondents within a limited time.

To maintain the privacy and comfort of the participants, the survey did not collect any personally identifiable information such as email addresses or other identification details that respondents might hesitate to share.

Using this method, a total of **66 responses** were collected, forming the dataset used for further preprocessing and exploratory data analysis. The number of responses satisfies the requirement mentioned in the assignment instructions, which specifies collecting more than 50 data samples.

4. DATASET OVERVIEW

The dataset used in this study was collected through an online survey created using Google Forms. The collected responses were exported as a spreadsheet and later converted into a CSV format for further data processing and analysis.

The raw dataset contains 66 responses, where each row represents an individual student who participated in the survey. The dataset consists of 19 variables, including a timestamp automatically generated by the Google Form at the time of submission.

The variables in the dataset capture different aspects of students' demographic information, classroom behavior, lifestyle habits, and academic outcomes. These include demographic attributes such as age and gender, behavioral factors such as seating preference, participation in class discussions, and attention level during lectures, and lifestyle-related variables such as sleep duration, sleep timing, screen usage, and phone usage during lectures.

Additionally, the dataset contains variables related to academic habits and performance, including average study hours per day, assignment preparation habits, time management skills, current CGPA, academic satisfaction, and overall academic stress level.

The dataset includes both categorical variables (for example, gender, seating preference, participation frequency, and assignment preparation timing) and numerical variables (such as attention level rating, study hours, phone usage duration, CGPA, and stress level rating). This combination of variable types allows for meaningful exploratory data analysis to investigate relationships between classroom behavior, lifestyle habits, and academic outcomes.

#	Column	Non-Null Count	Dtype
0	Timestamp	66 non-null	object
1	Age	66 non-null	int64
2	Gender	66 non-null	object
3	What best describes you?	66 non-null	object
4	Average Sleep hours per day	66 non-null	object
5	Usual sleep time	66 non-null	object
6	How would you rate your daytime alertness?	66 non-null	int64
7	Where do you usually prefer to sit in class?	66 non-null	object
8	How would you rate your attention level during lectures?	66 non-null	int64
9	How often do you participate in class discussions?	66 non-null	object
10	Average phone usage during lectures (minutes per class)	66 non-null	int64
11	Average screen time per day (in hours)	65 non-null	object
12	Do you use your phone while studying?	66 non-null	object
13	Average study hours per day	66 non-null	float64
14	When do you usually start working on assignments?	66 non-null	object
15	How would you rate your time management skills?	66 non-null	int64
16	Current CGPA (Not Mandatory)	58 non-null	float64
17	How satisfied are you with your academic performance?	66 non-null	int64
18	Overall academic stress level	65 non-null	float64

Figure 1. Summary of Raw Dataset

Some variables contain missing values. For example, the screen time variable contains one missing value, the CGPA variable contains eight missing values, and the academic stress level variable contains one missing value. These missing values are addressed during the data preprocessing stage before conducting exploratory data analysis.

A summary of the dataset structure, including column names, data types, and non-null counts, is shown in Figure 1.

5. DATA PREPROCESSING

After examining the dataset, I found that the dataset was not consistent and accurate because it contained NaN values, some inconsistent entries, unnecessary columns, and numerical columns stored with the object data type.

5.1. Removal of Unnecessary Columns

The timestamp column, which is automatically generated by Google Forms along with the dataset, is not useful for our study. Therefore, I decided to drop that column.

5.2. Column Name Cleaning

The original column names obtained from the survey contained long descriptions and extra spaces, which made them difficult to access during analysis. Therefore, I simplified the column names and converted them into concise and readable formats to enable easier referencing during the analysis process.

5.3. Handling Inconsistent Data Entries

Some of the responses contained textual or inconsistent values such as "6 hours", "6.5 hours", or ranges like "8–9 hours". These values were standardized by extracting the numeric portion of the responses. Some entries that could not be interpreted as valid numeric values were converted into missing values (NaN) to ensure uniform data representation.

5.4. Handling Missing Values

After handling the inconsistent data entries, we found that there were a total of 10 missing values: one from screen time, eight from CGPA, and the remaining one from sleep hours. Conditional mean imputation was used for the screen time column, while median imputation was applied to the remaining two columns.

5.5. Data Type Conversion

After cleaning the dataset, relevant columns were converted into appropriate data types. Numerical variables such as sleep hours, screen time, study hours, CGPA, and academic stress levels were converted into numeric data types to enable statistical analysis and visualization.

After completing these preprocessing steps, the dataset contained **66 observations and 18 variables with no missing values**, making it suitable for exploratory data analysis.

#	Column	Non-Null Count	Dtype
0	age	66 non-null	int64
1	gender	66 non-null	object
2	student_type	66 non-null	object
3	sleep_hours	66 non-null	float64
4	sleep_time	66 non-null	object
5	daytime_alertness	66 non-null	int64
6	seating_preference	66 non-null	object
7	attention_level	66 non-null	int64
8	class_participation	66 non-null	object
9	phone_usage_lecture	66 non-null	int64
10	screen_time	66 non-null	float64
11	phone_during_study	66 non-null	object
12	study_hours	66 non-null	float64
13	assignment_start_time	66 non-null	object
14	time_management	66 non-null	int64
15	cgpa	66 non-null	float64
16	academic_satisfaction	66 non-null	int64
17	academic_stress	66 non-null	float64

Figure 2. Summary of Cleaned Dataset

6. DESCRIPTIVE STATISTICS AND EXPLORATORY DATA ANALYSIS

6.1. Descriptive Analysis

Based on the descriptive analysis of the columns, I have understood the general characteristics and distribution of key variables in the dataset.

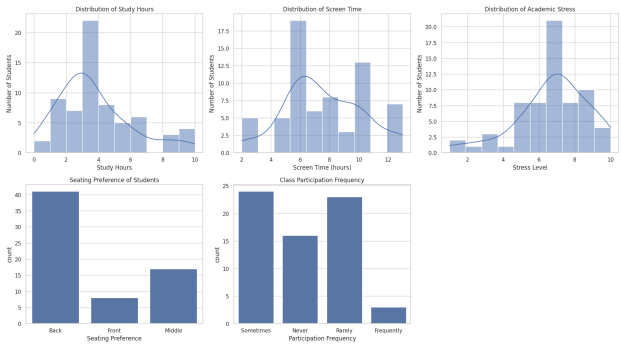

Figure 3. Frequency Distributions of some columns

- Study Hours: Most students study between 2–5 hours per day, indicating moderate study habits.
- Screen Time: Many students spend around 5–7 hours per day on digital devices, with a few reporting more than 10 hours.
- Academic Stress Levels: Most students report moderate to high stress levels, mainly between levels 6 and 8.
- Seating Preference: A large number of students prefer sitting at the back of the classroom, followed by the middle rows.
- Class Participation: Most students participate in discussions rarely or sometimes, while only a few participate frequently.

These observations provide a basic overview of student behavior, which will be further analyzed in the next sections to explore relationships between classroom behavior, lifestyle habits, and academic outcomes.

6.2. Research Question 1

Does seating preference and classroom participation influence academic performance and engagement?

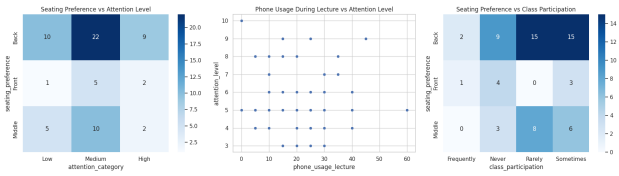

Figure 4. Plots for the Correlation and tests

6.2.1. *Seating Preference vs Attention Level*

Categorical split: Attention level → Low / Medium / High

- H0: Seating preference and attention level are independent
- H1: Seating preference and attention level are associated

Results

- Chi-square statistic: 1.53
- p-value: 0.821

Conclusion

- p-value > 0.05 → Fail to reject H0
- No statistically significant relationship between seating preference and attention level

6.2.2. *Class Participation vs CGPA*

Categorical split: CGPA → Low / Medium / High

- H0: Class participation and CGPA are independent
- H1: Class participation and CGPA are associated

Results

- Chi-square statistic: 5.14
- p-value: 0.526

Conclusion

- p-value > 0.05 → Fail to reject H0
- No statistically significant relationship between class participation and CGPA

6.2.3. *Phone Usage During Lectures vs Attention Level*

Both variables are numerical, so Pearson correlation was used instead of Chi-square.

Results

- Correlation coefficient: - 0.162

Conclusion

- Indicates a weak negative relationship
- Higher phone usage may slightly reduce attention, but the relationship is very weak

6.2.4. *Seating Preference vs Class Participation*

Both variables are categorical, so Chi-square test was applied.

Hypotheses

- H0: Seating preference and class participation are independent
- H1: Seating preference and class participation are associated

Results

- Chi-square statistic: 8.04
- p-value: 0.235

Conclusion

- p-value > 0.05 → Fail to reject H0
- No statistically significant relationship between seating preference and class participation

Overall Finding

Classroom behaviors such as seating preference, participation frequency, and phone usage during lectures show weak or statistically insignificant relationships with attention levels and academic performance in this dataset.

6.3. Research Question 2

How do sleep patterns, screen usage, and assignment management habits influence academic stress among students?

To analyze this research question, relationships between sleep hours, screen time, assignment start time, student type, and academic stress were examined. Correlation analysis was used for numerical variables, while Chi-square tests were used for categorical relationships.

6.3.1. *Sleep Hours vs Academic Stress*

- **Variables**: Sleep hours (numerical) and academic stress (numerical)
- **Method**: Pearson correlation analysis

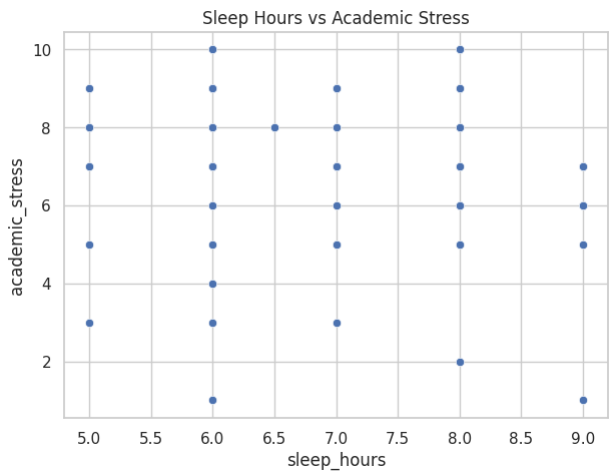

Figure 5. Sleep Hours vs Academic Stress

Result

- Correlation coefficient: -0.124

Conclusion

- Indicates a very weak negative relationship
- Students with more sleep tend to have slightly lower stress, but the relationship is not strong

6.3.2. Screen Time vs Academic Stress

- **Variables:** Screen time (numerical) and academic stress (numerical)
- **Method:** Pearson correlation analysis

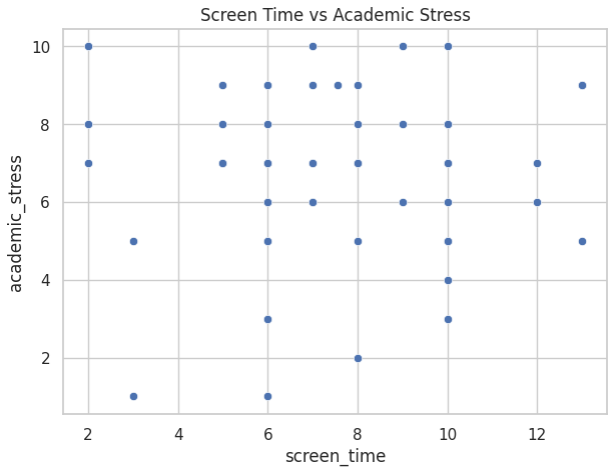

Figure 6. Screen Time vs Academic Stress

Result

- Correlation coefficient: 0.115

Conclusion

- Indicates a **very weak positive relationship**
- Higher screen time may slightly increase stress, but the relationship is **very weak**

6.3.3. Assignment Start Time vs Academic Stress

- **Variables:** Assignment start time (categorical) and academic stress (categorical grouping)
- **Method:** Chi-square test of independence

Hypotheses

- H0: Assignment start time and academic stress are independent
- H1: Assignment start time and academic stress are associated

Results

- Chi-square statistic: 6.23
- p-value: 0.398

Conclusion

- p-value > 0.05 → Fail to reject H0
- No statistically significant relationship between assignment start time and academic stress

6.3.4. Student Type vs Sleep Time

- **Variables:** Student type (Early Bird, Night Owl, Neutral) and sleep time
- **Method:** Chi-square test

Hypotheses

- H0: Student type and sleep time are independent
- H1: Student type and sleep time are associated

Results

- Chi-square statistic: 3.20
- p-value: 0.784

Conclusion

- p-value > 0.05 → Fail to reject H0
- No statistically significant relationship between student type and sleep time

Overall Finding

Lifestyle factors such as sleep hours, screen time, and assignment preparation habits show weak or statistically insignificant relationships with academic stress levels in this dataset.

7. KEY FINDINGS

From the exploratory data analysis, a few important observations were identified:

- Most students reported studying around 2-–5 hours per day, indicating moderate study habits.
- Many students spend 5–7 hours daily on digital devices, showing relatively high screen usage.
- Academic stress levels among students were generally moderate to high.
- Seating preference did not show a significant relationship with attention levels.
- Class participation did not have a clear statistical association with CGPA in this dataset.
- Phone usage during lectures showed a slight negative relationship with attention, but the effect was weak.
- Sleep hours and screen time showed only weak relationships with academic stress.
- Assignment start time also did not show a significant association with stress levels.

These findings summarize the main patterns observed from the dataset before drawing overall conclusions.

8. CONCLUSION

In this study, we explored how classroom behavior and lifestyle habits relate to students' academic engagement and stress levels. Based on the analysis, most of the factors examined—such as seating preference, class participation, sleep habits, screen usage, and assignment preparation—showed weak or statistically insignificant relationships with academic performance and stress levels in this dataset. Although some small patterns were observed, such as slightly lower attention with increased phone usage during lectures, the overall results suggest that these factors alone may not strongly determine students' academic outcomes.

However, this study has a few limitations. The dataset contains a relatively small sample size (66 responses), which may limit the generalizability of the results. In addition, the data was collected through self-reported surveys, which may introduce response bias. Furthermore, other factors that may influence academic performance and stress—such as teaching methods, course difficulty, personal motivation, and environmental conditions—were not included in this study.

A. APPENDIX

The dataset used in this study was collected through an online questionnaire created using Google Forms. The survey included questions related to students' classroom behavior, study habits, lifestyle patterns, and academic stress levels.

The Google Form used for collecting the responses can be accessed through the following link:

Survey Form Link:
https://forms.gle/wykQz6DPGB9zdT599